Type IV Equation of the form $z = px + qy + f(p,q)$
(Clairaut's equation)

$\frac{\partial z}{\partial x} = p$ and $\frac{\partial z}{\partial y} = q$

complete solution is given by $p = a$ and $q = b$

ie $z = ax + by + f(a,b)$

(1) solve $(p+q)(z - xp - yq) = 1$

$z - xp - yq = \frac{1}{p+q}$

$z = px + qy + \frac{1}{p+q}$

complete integral is given by

$z = ax + by + \frac{1}{a+b}$

(2) Find the complete integral of

$pqz = p^2(xq + p^2) + q^2(yp + q^2)$

$pqz = p^2xq + p^4 + q^2yp + q^4$

$z = px + \frac{p^3}{q} + qy + \frac{q^3}{p}$

ie $z = px + qy + \left(\frac{p^3}{q} + \frac{q^3}{p}\right)$

This is a Clairaut's diff. eq

complete integral is given by

$z = ax + by + \left(\frac{a^3}{b} + \frac{b^3}{a}\right)$

Find the complete integral of the following partial diff. eqs.

(i) $x^2p^2 + y^2q^2 = z^2$ (ii) $z(p^2 - q^2) = x - y$

(i) $x^2p^2 + y^2q^2 = z^2$

i.e. $\left(\frac{x^2p^2}{z^2}\right) + \left(\frac{y^2q^2}{z^2}\right) = 1$

$\left(\frac{xp}{z}\right)^2 + \left(\frac{yq}{z}\right)^2 = 1$

$p = \frac{\partial z}{\partial x}, \quad q = \frac{\partial z}{\partial y}$

i.e. $\left(\frac{x}{z}\frac{\partial z}{\partial x}\right)^2 + \left(\frac{y}{z}\frac{\partial z}{\partial y}\right)^2 = 1 \longrightarrow (1)$

Put $\frac{dx}{x} = dX, \quad \frac{dy}{y} = dY, \quad \frac{dz}{z} = dZ$

Eq (1) can be written as

$\left(\frac{\partial z}{z} / \frac{\partial x}{x}\right)^2 + \left(\frac{\partial z}{z} / \frac{\partial y}{y}\right)^2 = 1$

i.e. $\left(\frac{dZ}{dX}\right)^2 + \left(\frac{\partial Z}{\partial Y}\right)^2 = 1$

i.e. $P^2 + Q^2 = 1$ where $P = \frac{\partial Z}{\partial X}, \quad Q = \frac{\partial Z}{\partial Y}$

This is a type I equation

solution is given by

$u = aX + bY + c$ where $a^2 + b^2 = 1$

$a^2 + b^2 = 1 \implies b^2 = 1 - a^2$

$\implies b = \sqrt{1-a^2}$

complete integral is given by

$Z = aX + \sqrt{1-a^2}\, Y + c$

i.e. $\log z = a \log x + \sqrt{1-a^2} \log y + \log b$

$= \log x^a + \log y^{\sqrt{1-a^2}} + \log b$

$= \log x^a y^{\sqrt{1-a^2}} b$

i.e. $z = b\, x^a y^{\sqrt{1-a^2}}$

(ix) $z(p^2-q^2) = x-y$

$$zp^2 - zq^2 = x-y$$

$$(\sqrt{z}\,p)^2 - (\sqrt{z}\,q)^2 = x-y$$

ie $\left(\sqrt{z}\,\frac{\partial z}{\partial x}\right)^2 - \left(\sqrt{z}\,\frac{\partial z}{\partial y}\right)^2 = x-y$

put $\sqrt{z}\,dz = dZ$ ie $Z = \frac{2}{3} z^{3/2}$

$$\therefore \left(\frac{dZ}{dx}\right)^2 - \left(\frac{dZ}{dy}\right)^2 = x-y$$

ie $P^2 - Q^2 = x-y$ where $P = \frac{\partial Z}{\partial x}$, $Q = \frac{\partial Z}{\partial y}$

ie $P^2 - x = Q^2 - y$

This is a type III equation

$$\therefore P^2 - x = Q^2 - y = a \text{ (say)}$$

$$P^2 - x = a, \quad Q^2 - y = a$$

$$P^2 = x+a, \quad Q^2 = y+a$$

$$P = \sqrt{x+a}, \quad Q = \sqrt{y+a}$$

$$dZ = P\,dx + Q\,dy$$

$$= \sqrt{x+a}\,dx + \sqrt{y+a}\,dy$$

integrating

$$Z = \frac{(x+a)^{3/2}}{3/2} + \frac{(y+a)^{3/2}}{3/2} + \frac{b}{3/2}$$

ie $\frac{2}{3} z^{3/2} = \frac{2}{3}\left[(x+a)^{3/2} + (y+a)^{3/2} + b\right]$

ie $z^{3/2} = (x+a)^{3/2} + (y+a)^{3/2} + b$

is the required complete integral

• Method of separation of variables

In this method we assume that given partial diff. eq has a solution of the form $z = X(x)\,Y(y)$ where X is a function of x and Y is a function of y

$z = XY$ $\qquad \frac{\partial z}{\partial x} = \frac{dX}{dx}Y$, $\frac{dz}{dy} = \frac{dY}{dy}X$

$\frac{\partial z}{\partial x} = X'Y$, $\frac{\partial z}{\partial y} = XY'$

substituting this in the given diff. we try to solve the given diff. eq.

Example (1)

Solve the equation $\frac{\partial u}{\partial x} = 4\frac{\partial u}{\partial y}$ where $u(0,y) = 8e^{-3y}$ by the method of separation of variables.

Let us assume a solution of the form

$u = XY$ where X is a function of x and Y is a function of y

$\frac{\partial u}{\partial x} = X'Y$, $\frac{\partial u}{\partial y} = XY'$

substituting in the given diff. eq

$X'Y = 4XY'$, Dividing by XY

$\frac{X'}{X} = \frac{4Y'}{Y} = K$ (say)

$X' = KX \longrightarrow (1)$ $\qquad 4Y' = KY \longrightarrow (2)$

i.e. $\frac{dX}{dx} = KX$ and $4\frac{dY}{dy} = KY$

i.e. $\frac{dX}{X} = K\,dx$ Integrating $\log X = Kx + \log a$

ie $\log\left(\frac{X}{a}\right) = Kx \Rightarrow \frac{X}{a} = e^{Kx}$

$X = a e^{Kx}$

From eq (2) $4\frac{dY}{dy} = KY \Rightarrow \frac{dY}{Y} = \frac{K}{4}dy$

Integrating $\log Y = \frac{K}{4}y + \log b$

ie $\log\frac{Y}{b} = \frac{KY}{4}$

$\Rightarrow \frac{Y}{b} = e^{\frac{KY}{4}}$ ie $Y = b e^{\frac{KY}{4}}$

$\therefore u = (a e^{Kx})(b e^{\frac{Ky}{4}})$ $(\because u = XY)$

$u = ab\, e^{Kx + \frac{Ky}{4}}$

ie $u = ab\, e^{K(x + \frac{y}{4})}$

It is given that $u(0,y) = 8e^{-3y}$ $\rightarrow (3)$

But $u(0,y) = ab\, e^{\frac{Ky}{4}}$ $\rightarrow (4)$

From (3) and (4) $ab = 8$ $\frac{K}{4} = -3$, $K = -12$

$\therefore u = 8e^{-12(x + \frac{y}{4})}$, $u = e^{-(12x+3y)}$

Example (2)

Using method of separation variables

Solve $\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u$, given $u(x,0) = 6e^{-3x}$

Let us assume a solution of the form

$u = XY$ where X is a function of x

and Y is a function of y

$\frac{\partial u}{\partial x} = X'Y$, $\frac{\partial u}{\partial y} = XY'$

Sub. in the given diff. eq

$X'Y = 2XY' + XY$, Dividing by XY

$\frac{X'}{X} = 2\frac{Y'}{Y} + 1$

$$\frac{X'}{X} = 2\frac{Y'}{Y} + 1 = K \text{ (say)}$$

$X' = KX \longrightarrow (1)$ $\quad \frac{2Y'}{Y} = K-1 \longrightarrow (2)$

From eq (1) $\frac{dX}{dx} = KX$

i.e. $\frac{dX}{X} = K\,dx$ Integrating

$\log X = Kx + \log a$

$\log \frac{X}{a} = Kx \Rightarrow \frac{X}{a} = e^{Kx}$

i.e. $X = a e^{Kx}$

From eq (2) $2Y' = (K-1)Y$

i.e. $\frac{dY}{dy} = \frac{K-1}{2} Y$

i.e. $\frac{dY}{Y} = \frac{K-1}{2} dy$

Integrating $\log Y = \frac{K-1}{2} y + \log b$

$\therefore \log \frac{Y}{b} = \frac{K-1}{2} y \Rightarrow \frac{Y}{b} = e^{\frac{K-1}{2}y}$

i.e. $Y = b e^{\frac{K-1}{2}y}$

$\therefore u = ab\, e^{Kx} \cdot e^{\frac{K-1}{2}y}$ $\quad (\because u = XY)$

i.e. $u = ab\, e^{Kx + \frac{K-1}{2}y}$

Given that $u(x,0) = 6e^{-3x} \longrightarrow (3)$

But $u(x,0) = ab\, e^{Kx} \longrightarrow (4)$

From (3) and (4) $ab = 6$ and $K = -3$

$\therefore$ Required solution is given by

$u = 6\, e^{-3x-2y}$

i.e. $u = 6\, e^{-(3x+2y)}$

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Partial diff. eqs. of second and higher order

In this sectn we discuss homogeneous ~~d~~linear ~~diff~~ diff. eqs with constant coeffts which are of the form

$$\frac{\partial^n z}{\partial x^n} + A_1 \frac{\partial^n z}{\partial x^{n-1}.\partial y} + A_2 \frac{\partial^n z}{\partial x^{n-2}\partial y^2} + \cdots + A_n \frac{\partial^n z}{\partial y^n} = f(x,y)$$

Denoting $\frac{\partial}{\partial x}$ by D and $\frac{\partial}{\partial y}$ by D', it can be written as

$$(D^n + A_1 D^{n-1} D' + A_2 D^{n-2} D'^2 + \cdots + A_n D'^n) z = f(x,y)$$

or $F(D D') = f(x,y) \longrightarrow (1)$

where $F(D, D') = D^n + A_1 D^{n-1} D' + A_2 D^{n-2} D'^2 + \cdots + A_n D'^n$

Solution of linear partial diff. eq.

As in the case of ordinary linear diff. eq. the complete integral of (1) consists of two parts

(a) complementary function (CF)

(b) Particular integral (PI)

where C.F is the general solution of (Assuming RHS as 0)

$F(D D')z = 0 \longrightarrow (2)$

Let $z = F(y+mx)$ be a soln of this eq

$Dz = \frac{\partial z}{\partial x} = m F'(y+mx)$, $D^2 z = \frac{\partial^2 z}{\partial x^2} = m^2 F''(y+mx)$

$\cdots \quad D^n z = \frac{\partial^n z}{\partial x^n} = m^n F^n (y+mx)$

Also $D' z = \frac{\partial z}{\partial y} = F'(y+mx)$,

$D'^2 z = \frac{\partial^2 z}{\partial y^2} = F''(y+mx) \cdots \quad D'^n z = \frac{\partial^n z}{\partial y^n} = F^{(n)}(y+mx)$

Thus $D^r D'^{n-r} z = \frac{\partial^n z}{\partial x^r \partial y^{n-r}} = \frac{\partial^r}{\partial x^r}\left(\frac{\partial^{n-r} z}{\partial y^{n-r}}\right)$

$= m^r F^n(y+mx)$

sub. in eq (2) we get

$F(D,D')z = (m^n + A_1 m^{n-1} + A_2 m^{n-2} + \dots + A_n) F^n(y+mx)$

$F(D,D')z = 0 \Longrightarrow$

$m^n + A_1 m^{n-1} + A_2 m^{n-2} + \dots + A_n = 0$

This is called the auxillary equation

If $m_1, m_2 \dots m_n$ are distinct roots of the auxillary equation then CF of eq (1) is

$z = \phi_1(y+m_1x) + \phi_2(y+m_2x) + \dots + \phi_n(y+m_nx)$

where $\phi_1, \phi_2 \dots \phi_n$ are arbitrary functions

Note The auxillary eq is obtained by putting $D=m$ and $D'=1$ in $F(D,D')=0$

when the auxillary eq has repeated roots we can modify the part of CF corresponding to these roots

If m is a double root corresponding part of CF canbe modified as

$z = x f_1(y+mx) + f_2(y+mx)$

If m is a root repeated r times corresponding part of CF can be written as

$z = x^{r-1} f_1(y+mx) + x^{r-2} f_2(y+mx) + \dots f_r(y+mx)$

(1) Solve $2r + 5s + 2t = 0$

$$r = \frac{\partial^2 z}{\partial x^2} = D^2 z, \quad s = \frac{\partial^2 z}{\partial x \partial y} = DD'z, \quad t = \frac{\partial^2 z}{\partial y^2} = D'^2 z$$

$\therefore$ Given eq can be written as

$$(2D^2 + 5DD' + 2D'^2)z = 0$$

A.E. is $2m^2 + 5m + 2 = 0$ (Put $D = m$, $D' = 1$)

ie $2m^2 + 4m + m + 2 = 0$

$2m(m+2) + (m+2) = 0$

ie $(m+2)(2m+1) = 0$

$\therefore m = -2, -\frac{1}{2}$ are two distinct roots

$\therefore z = f_1(y - 2x) + f_2\left(y - \frac{x}{2}\right)$

ie $z = f_1(y - 2x) + \phi_1(2y - x)$ is the gen. soln.

(2) Solve $r = a^2 t$

$r = D^2 z$, $t = D'^2 z$

$(D^2 - a^2 D'^2)z = 0$

Auxillary eq is $m^2 - a^2 = 0$

ie $(m+a)(m-a) = 0$ ie $m = -a, a$

Hence the solution is

$y = f_1(y + ax) + f_2(y - ax)$

(3) Solve $\dfrac{\partial^3 z}{\partial x^3} - 3\dfrac{\partial^3 z}{\partial x^2 \partial y^2} + 2\dfrac{\partial^3 z}{\partial x \partial y^2} = 0$

Given eq. can be written as

$$(D^3 - 3D^2 D' + 2DD'^2)z = 0$$

Auxillary eq is given by

$m^3 - 3m^2 + 2m = 0$

ie $m(m^2-3m+2)=0$

or $m(m-1)(m-2)=0$

$m=0,1,2$

Hence general soln is given by

$$z=f_1(y)+f_2(y+x)+f_3(y+2x)$$

solve $(D^3-4D^2D'+4DD'^2)z=0$

Auxillary eq is given by

$$m^3-4m^2+4m=0$$

$$m(m^2-4m+1)=0$$

$$m(m-2)^2=0$$

$\therefore m=0,2,2$

Hence the solution is

$$z=f_1(y)+f_2(y+2x)+xf_3(y+2x).$$

solve $\frac{\partial^4 z}{\partial x^4}-\frac{\partial^4 z}{\partial y^4}=0$

ie $(D^4-D'^4)z=0$

A.E. is $m^4-1=0$

ie $(m^2-1)(m^2+1)=0$

ie $(m-1)(m+1)(m+i)(m-i)=0$

ie $m=1,-1,i,-i$

Hence solution is given by

$$z=f_1(y+x)+f_2(y-x)+f_3(y+ix)+f_4(y-ix)$$

Solve the following p.d.eqs

(1) $25r-40s+16t=0$

(2) $(D^3-6D^2D'+11DD'^2-6D'^3)z=0.$

consider the diff. eq $f(D D') z = \phi(x, y)$
General solution of this p.d. eq is given by

$$z = CF + PI$$

Particular integral may be written as

$$PI = \frac{1}{f(D, D')} \phi(x, y)$$

Factorizing it, resolving into partial fractions or expanding as an infinite series we try to find the PI.

Case I $\phi(x, y)$ is a polynomial

Example solve $(D^2 - DD' + D'^2) z = 12xy$

A. E is given by $m^2 - 2m + 1 = 0$

or $(m-1)^2 = 0$ $\therefore m = 1, 1$

$CF = f_1(y + x) + x f_2(y + x)$

Now $PI = \frac{1}{D^2 - DD' + D'^2} (12xy)$

$$= \frac{1}{(D - D')^2} 12xy$$

$$= \frac{1}{D^2 \left(1 - \frac{D'}{D}\right)^2} 12xy$$

$$= \frac{1}{D^2} \left(1 - \frac{D'}{D}\right)^{-2} 12xy$$

$$= \frac{1}{D^2} \left[1 + 2\frac{D'}{D} + 3\frac{D'^2}{D^2} + \cdots\right] 12xy$$

$(\because (1-x)^{-2} = 1 + 2x + 3x^2 + \cdots)$

$$= \frac{1}{D^2} \left[12xy + \frac{2}{D}(12x) + 0\right]$$

$$= 12 \frac{1}{D^2} (xy) + 24 \frac{1}{D^3} (x)$$

$$\boxed{\frac{1}{D}(xy) = \frac{x^2}{2} y \qquad \frac{1}{D^2}(xy) = \frac{x^3}{6} y}$$

$$= 12 \left(\frac{x^3 y}{6}\right) + 24 \left(\frac{x^4}{24}\right) = 2x^3 y + x^4$$

gen soln $z = CF + PI = f_1(y + x) + x f_2(y + x) + 2x^3 y + x^4$

(2) Solve $(D^2 + 3DD' + 2D'^2)z = x+y$

A.E is $m^2 + 3m + 2 = 0$ i.e $(m+1)(m+2) = 0$

$\therefore m = 1, -2$

$CF = f_1(y-x) + f_2(y-2x)$

Now $PI = \dfrac{1}{D^2+3DD'+2D'^2}(x+y)$

$= \dfrac{1}{D^2\left(1+\frac{3D'}{D}+\frac{2D'^2}{D^2}\right)}(x+y)$

$= \dfrac{1}{D^2}\left(1+\frac{3D'}{D}+\frac{2D'^2}{D^2}\right)^{-1}(x+y)$

$= \dfrac{1}{D^2}\left[1-\left(\frac{3D'}{D}+\frac{2D'^2}{D^2}\right)+\cdots\right](x+y)$

$= \dfrac{1}{D^2}\left[1-\frac{3D'}{D}\right](x+y)$ $\left(D'(x+y) = 0+1 = 1\right)$

$= \dfrac{1}{D^2}(x+y) - \dfrac{3}{D^3}(1)$

$= \dfrac{x^3}{6} + \dfrac{yx^2}{2} - 3\dfrac{x^3}{6}$

$= \left(\dfrac{x^3}{6} - \dfrac{3x^3}{6}\right) + \dfrac{x^2y}{2}$

$= -\dfrac{1}{3}x^3 + \dfrac{x^2y}{2}$

General solution is

$y = CF + PI = f_1(y-x) + f_2(y-2x) - \dfrac{1}{3}x^3 + \dfrac{x^2y}{2}$